Incidence operators and codes: manuscript lengths

Whole common theory, two complete applications, and declared proof inputs

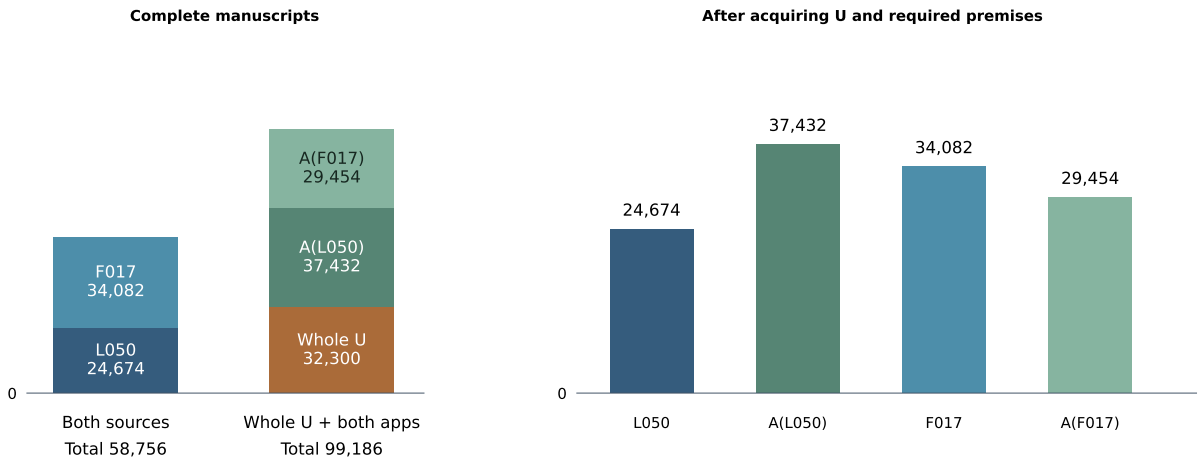
The two complete sources contain 58 756 nonspace characters in the fixed primary coordinate. The entire common paper and both complete applications contain 99 186. Thus the whole manuscript factor is

$$\frac{c(L050)+c(F017)}{c(U)+c(A_{L050})+c(A_{F017})}=\frac{24\,674+34\,082}{32\,300+37\,432+29\,454}=\boxed{0.592382}.$$

This is expansion: the complete replacement is longer. There is no achieved whole-topic compression in this coordinate. The factor is source length divided by replacement length; a factor below one means expansion.

The primary coordinate is NFC nonspace characters in the accepted calibrated selected extraction. Whole manuscripts include titles, abstracts, tables, proof and algorithm prose, footnotes and references. Only precisely identified running furniture is removed. There are no executable listings in these five PDFs, so symmetric listing exclusion removes no passages here.

Complete document	Physical pages	Primary length
L050	11	24 674
F017	17	34 082
Whole U	14	32 300
A(L050)	15	37 432
A(F017)	12	29 454



NFC nonspace characters, calibrated selected text. Both panels start at zero; their scales differ.

After acquiring the *whole* common paper, the complete declared required program/data/record collection, and the actual retained mathematical premises, the conditional comparisons are

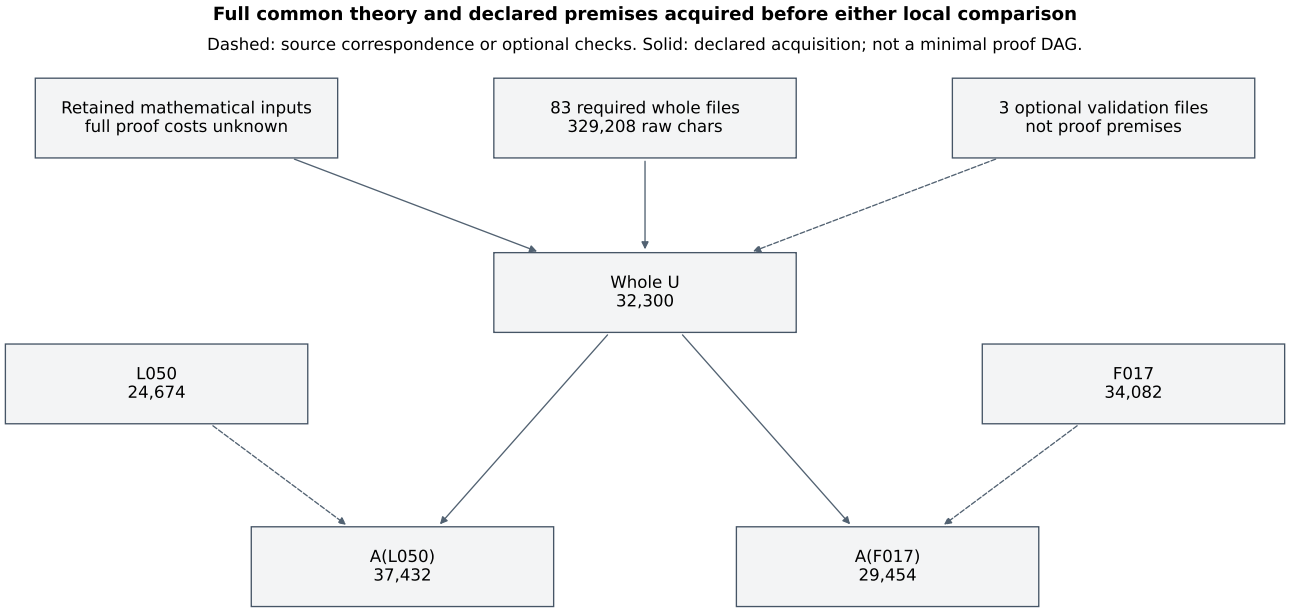
$$\frac{c(L050)}{c(A_{L050})}=\frac{24\,674}{37\,432}=\boxed{0.659169},\qquad \frac{c(F017)}{c(A_{F017})}=\frac{34\,082}{29\,454}=\boxed{1.157126}.$$

The L050 application is longer than its source; the F017 application is shorter only in this conditional comparison. Neither comparison discards the initial acquisition or substitutes for the whole-topic result. No sibling application is a proof premise. Optional validation is not a required premise.

L050: D. Leemans and B. G. Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, Mathematical Communications 21 (2016), 239–249. F017: J. D. Key and B. G. Rodrigues, *LCD codes from adjacency matrices of graphs*, complete seventeen-page author manuscript dated 4 August 2017.

What must be acquired

The entire fourteen-page common paper is paid once, including interfaces unused by a particular local application. The fifteen-page L050 and twelve-page F017 applications are both paid in the whole comparison. In the displayed order, cumulative manuscript costs are 32 300; 69 732; 99 186 (whole common; common plus L050; common plus both applications). This order is a cost accounting convention, not a claim about chronological discovery or a minimal proof dependency graph.



The conditional convention conservatively acquires the *complete topic-wide* required union: 83 unique whole files occupying 422 956 original bytes. It includes 20 programs, two generated certificates, 19 mandatory data files and 42 retained finite records. No claim is made that every one of these files is necessary for each individual application. Required computations are not relabelled as optional, and completed enumeration records remain premises. The numerical replay does not rerun those enumerations.

Whole-file role	Files	Original bytes	NFC nonspace chars
Required programs	20	109 381	87 327
Derived certificates	2	18 313	13 234
Mandatory data	19	131 870	117 347
Finite records	42	163 392	111 300
Optional validation	3	11 792	7 896
Operational drivers	11	103 622	91 772
Partial Lean files	2	6 758	4 726
Partial scope files	3	3 122	2 569
All unique complete contents	102	548 250	436 171

All 127 occurrences are disclosed; exactly byte-identical complete contents are counted once within this topic. The three optional validation files are two programs and an auxiliary JSON file. Eleven operational drivers and five partial-formal files are separate from mathematical premises. The two Lean source files and three scope files concern the same nine generic declarations; they do not formalize both complete sources.

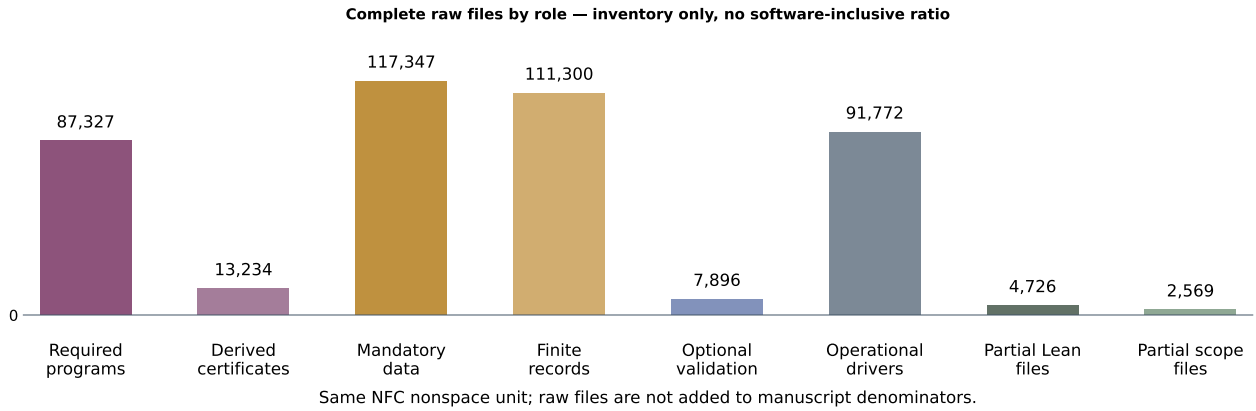
Historical source software, full retained classical proofs, graphics/layout, external libraries and complete formalization have unknown costs. These are *unknown, not zero.* The raw inventory is not added to a manuscript comparison and no software-inclusive factor is reported. Full paths, hashes, roles, duplicate occurrences and all ten raw-file metric coordinates are available in the accompanying JSON and CSV files.

Sensitivity and limits

All thirty fixed policy coordinates are retained: three extraction variants, NFC and NFKC, and five units. Each has one whole manuscript fraction and two conditional fractions, stored as exact integers and reduced fractions. The following five-unit table uses calibrated selected NFC text.

Unit	Whole	L050 conditional	F017 conditional
Whitespace tokens	0.737219	0.886729	1.263282
Alphanumeric units	0.726938	0.808612	1.404554
Nonspace characters	0.592382	0.659169	1.157126
Collapsed characters	0.617208	0.694098	1.178004
Collapsed UTF-8 bytes	0.633315	0.704043	1.220450

Nonspace-character variant	Whole	L050 conditional	F017 conditional
raw untrimmed, NFC	0.599087	0.668117	1.168380
raw untrimmed, NFKC	0.600032	0.669105	1.170076
raw selected, NFC	0.592382	0.659169	1.157126
raw selected, NFKC	0.593328	0.660157	1.158827
calibrated selected, NFC	0.592382	0.659169	1.157126
calibrated selected, NFKC	0.593328	0.660157	1.158827



Raw untrimmed text is the literal PDF extractor output, including furniture; raw selected text removes only the accepted furniture; calibrated selected text also makes the accepted occurrence-bound glyph repairs. Printed hyphens remain. The accepted audit binds actual font dictionaries and events, and preserves the explicit Type0 extraction qualification. These are extracted-text coordinates, not full visual or graphic-complexity measurements. Physical extent remains 28 source pages and 41 replacement pages.

The applications recover the *corrected* accepted contracts: 54 L050 and 104 F017 dispositions. Named graph identifications, classification/Brauer inputs, earlier code results and the seven declared F017 input classes remain attributed and retained. Four large Paley pairs retain source upper bounds. The source questions about all $\mathrm{SRG}(253, 140, 87, 65)$, the other fifteen-dimensional modules, general further families and efficient decoding are not newly solved here. The source maps preserve the exact corrections and qualifications; comparing lengths does not validate a false literal source statement.

This numerical edition reuses the separately accepted scientific and input packets. It supplies no new original, question or formalization credit. Its checks reproduce the integer arithmetic, complete inventory, tables and actual vector bars; they do not replace the underlying mathematical reviews.